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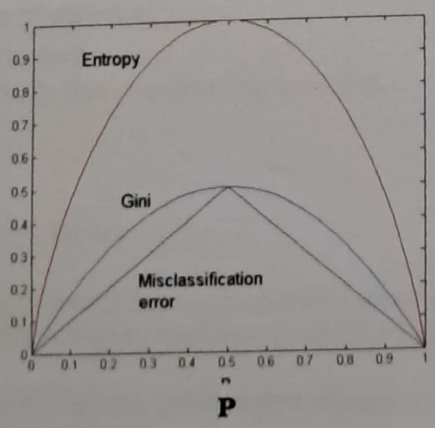
Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Date	26 November 2024	Maximum Marks	60
Course Code	IS353IA	Duration	120 min
Sem	V	CIE-1	
UG/PG	UG	Faculty:	

Artificial Intelligence and Machine Learning
<(Common to AIML/CSE/ISE) >

Sl. No.	Quiz Questions	M	L	CO
1	Justify whether the following sentence(s) hold true or false for Turing test in context AI. Marks are awarded for justification only. i. The Turing Test requires the machine to be able to convincingly simulate human intelligence to pass. *TRUE- The Turing Test, proposed by Alan Turing, evaluates a machine's ability to exhibit behavior indistinguishable from that of a human during a conversation.* ii. The Turing Test guarantees that a machine that passes it is intelligent. *FALSE- The Turing Test only measures a machine's ability to mimic human-like responses but does not guarantee genuine intelligence or understanding.*	02	L3	CO1
2	The _____ algorithm uses Entropy as a measure for constructing decision trees, while the _____ algorithm employs the Gini Index for splitting criteria. *ID3, CART*	02	L1	CO1
3	Draw observations from the graph below  At p=0.5, all measures are at their maximum (highest impurity).	02	L3	CO1

1a - 8
1b 6
14

2a - 12
2b - 6
18

3a - 10
3b - 8
18.

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	• As p approaches 0 or 1 (classes become more pure), all measures decrease. • Entropy drops faster than Gini, and Misclassification Error drops the slowest, reflecting their sensitivity to changes in p.			
4	List the key stopping criteria used in constructing a decision tree (any two) • *Maximum depth reached* *No significant information gain* *Pure node (homogeneous labels)* *all attribute same value*	**02**	L2	CO1
5	Differentiate Rationality and perfection in AI decision making with an example. *Rationality: Making the best decision given the available information, computational limitations, and environmental constraints.* *Perfection: Making an ideal decision with complete and perfect information and unlimited computational power.*	**02**	L2	CO1

Sl. No.	Test Questions	M	L	CO
1a	Analyze each of the following agent environments and determine whether they are fully or partially observable, deterministic or stochastic, static or dynamic, and discrete or continuous. Justify your classification with appropriate reasoning for each environment. i. autonomous Mars rover. ii. playing tic-tac-toe. iii. mathematician's theorem-proving assistant. **2.5*2+3=8M** **i. Autonomous Mars Rover** • **Observability:** Partially Observable (limited sensor range and obstructions). • **Deterministic/Stochastic:** Stochastic (unpredictable elements like dust storms). • **Static/Dynamic:** Dynamic (environment changes over time).	**08**	L3	CO1

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- **Discrete/Continuous:** Continuous (real-world terrain and movements).

ii. Playing Tic-Tac-Toe

- **Observability:** Fully Observable (entire board is visible).
- **Deterministic/Stochastic:** Deterministic (outcomes depend solely on player moves).
- **Static/Dynamic:** Static (game state doesn't change without player action).
- **Discrete/Continuous:** Discrete (finite moves and board states).

iii. Mathematician's Theorem-Proving Assistant

- **Observability:** Fully Observable (complete problem description available).
- **Deterministic/Stochastic:** Deterministic (logical rules ensure predictable outcomes).
- **Static/Dynamic:** Static (problem doesn't change during solving).
- **Discrete/Continuous:** Discrete (steps involve logical, finite reasoning).

1b — **06** — L3 — CO1

For the following examples, find the PEAS (Performance measure, Environment, Actuators, Sensors) framework and describe each component:

1. Washing Machine
2. Autonomous Drone
3. Smart Home Thermostat

2.5*2+3=8M

Example	Performance Measure	Environment	Actuators	Sensors
Washing Machine	Cleanliness of clothes, minimal water and energy usage, adherence to time settings	Laundry load, type of clothes, water supply, detergent	Drum motor, water inlet valve, detergent dispenser, heater	Water level sensor, temperature sensor, load sensor, speed sensor
Autonomous Drone	Safe navigation, task completion (e.g., delivery, surveillance), energy efficiency	Airspace, obstacles, GPS location, weather conditions	Motors, rotors, camera gimbals, control surfaces	GPS, gyroscope, accelerometer, camera, lidar, altimeter
Smart Home Thermostat	Maintaining desired	Room temperature,	Heating or cooling	Temperature sensor, humidity

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	temperature with minimal energy consumption	external weather, occupancy, HVAC system	system control, fan control	sensor, occupancy sensor			

4.

Consider the following data set:

price	maintenance	capacity	airbag	profitable
low	low	2	no	yes
low	med	4	yes	no
low	low	4	no	yes
low	high	4	no	no
med	med	4	no	no
med	med	4	yes	yes
med	high	2	yes	no
med	high	5	no	yes
high	med	4	yes	yes
high	high	2	yes	no
high	high	5	yes	yes

3a Suppose we decide to construct a decision tree using binary splits and the Gini index impurity measure. Which feature and split point combinations would be the best to use as the root node assuming that we consider each of the input features to be unordered? **12** L3 CO2

Feature	P	Gini Index
price	price = low	0.3961
maintenance	maintenance = low	0.3961
capacity	capacity = 4	0.3879
airbag	airbag = no	0.4909

Capacity is best feature

3b Discuss the approaches for splitting based on continuous attributes? Demonstrate the same using the following dataset: 3+3=6M **06** L3 CO2

Record	Age	Purchased (Yes/No)
1	22	No
2	25	Yes
3	28	Yes

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4	35	No
5	40	Yes

Binary split:

Threshold = 28: Split into Age ≤ 28 and Age > 28.

- Age ≤ 28: Records 1, 2, 3 (Yes: 2, No: 1)
- Age > 28: Records 4, 5 (Yes: 2, No: 0)
- **Gini for Age ≤ 28:**

$$Gini = 1 - \left(\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2\right) = 0.4444$$

- **Gini for Age > 28:**

$$Gini = 1 - \left(\left(\frac{2}{2}\right)^2 + \left(\frac{0}{2}\right)^2\right) = 0$$

- **Weighted Gini for Threshold = 28:**

$$Gini_{split} = \frac{3}{5} \times 0.4444 + \frac{2}{5} \times 0 = 0.2667$$

Discrtization approach

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Step 2: Calculate Gini Index for Each Interval

1. **Interval 1 ($\text{Age} \leq 25$):**
   - Records: 1, 2, 3
   - Purchased: Yes = 2, No = 1
   - Gini Index:

$$Gini = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = 1 - 0.4444 - 0.1111 = 0.4444$$

2. **Interval 2 ($25 < \text{Age} \leq 35$):**
   - Record: 4
   - Purchased: Yes = 0, No = 1
   - Gini Index:

$$Gini = 1 - \left(\frac{0}{1}\right)^2 - \left(\frac{1}{1}\right)^2 = 0$$

3. **Interval 3 ($\text{Age} > 35$):**
   - Record: 5
   - Purchased: Yes = 1, No = 0
   - Gini Index:

$$Gini = 1 - \left(\frac{1}{1}\right)^2 - \left(\frac{0}{1}\right)^2 = 0$$

4a Compare and contrast the types of intelligent agents—simple reflex, model-based reflex, goal-based, and utility-based—based on their rationality, memory use, ability to handle dynamic or partially observable environments, and computational complexity. Use examples like a robot vacuum or a virtual assistant to illustrate their differences and applications. Present your answer in a tabular format with the following columns: 2*5=10M **10** L3 CO1

Agent Type	Rationality	Memory Usage	Handling Dynamic/Partially Observable Environments	Computational Complexity	Example/Use Case
Simple Reflex Agent	Acts solely based on current percepts without	Minimal	Cannot handle dynamic or partially observable environments effectively	Low (simple condition-action rules)	Robot vacuum cleaning a static, fully observable

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		considering history or future; rational in simple tasks				room			
	Model-Based Reflex Agent	Rational with the inclusion of an internal model to track the environment state	Moderate (needs to store model)	Can handle partially observable environments to some extent	Moderate (requires model maintenance)	Robot vacuum navigating around moving obstacles			
	Goal-Based Agent	Rational by choosing actions aimed at achieving defined goals	High (stores model and goals)	Can handle dynamic and partially observable environments with goal evaluation	High (requires planning and search algorithms)	Virtual assistant scheduling appointments based on user goals			
	Utility-Based Agent	Rational by selecting actions that maximize utility or satisfaction	Very High (tracks model, goals, and utility)	Best suited for complex, dynamic, and partially observable environments	Very High (requires optimization algorithms)	Self-driving car ensuring safety, efficiency, and passenger comfort			
4b 3	With the help of examples describe the potential causes and solutions of model overfitting in decision trees. 1. Overfitting Due to Presence of Noise 2. Overfitting Due to Lack of Representative Samples 3. High model complexity (Multiple Comparison Procedure) 2.5*2+3=8M						**08**	L2	CO1

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BT-Blooms Taxonomy, CO-Course Outcomes

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	42	18				2	12	46			

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Gini Index

I

$$Gini(Price(low)) = 1-\left(\frac{2}{4}\right)^2-\left(\frac{2}{4}\right)^2 = 1-0.25-0.25 = 0.5$$

$$Gini(Price(med)) = 1-\left(\frac{2}{4}\right)^2-\left(\frac{2}{4}\right)^2 = 0.5$$

$$Gini(Price(high)) = 1-\left(\frac{2}{3}\right)^2-\left(\frac{1}{3}\right)^2 = 1-0.44-0.11 = 0.45$$

$$Gini\ Index(Price) = \frac{4}{11}*0.5 + \frac{4}{11}*0.5 + \frac{3}{11}*0.45$$

$$= 0.181 + 0.181 + 0.1227$$

$$= \underline{0.484}$$

II

$$Gini(Maintainance(low)) = 1-\left(\frac{2}{2}\right)^2-\left(\frac{0}{2}\right)^2 = 0$$

$$Gini(Maintainance(med)) = 1-\left(\frac{2}{4}\right)^2-\left(\frac{2}{4}\right)^2 = 0.5$$

$$Gini(Maintainance(high)) = 1-\left(\frac{3}{5}\right)^2-\left(\frac{2}{5}\right)^2$$

$$= 1-0.36-0.16 = 0.48$$

$$Gini\ Index(Maintainance) = \frac{2}{11}*0 + \frac{4}{11}*0.5 + \frac{5}{11}*0.48$$

$$= 0.181 + 0.218 = \underline{0.399}$$

III.

$$Gini(\text{airbag}(yes)) = 1-\left(\frac{3}{6}\right)^2-\left(\frac{3}{6}\right)^2 = 0.5$$

$$Gini(airbag(no)) = 1-\left(\frac{3}{5}\right)^2-\left(\frac{2}{5}\right)^2 = 0.48$$

$$Gini(airbag) = \frac{6}{11}*0.5 + \frac{5}{11}*0.48 =$$

$$= 0.27 + 0.218 = \underline{0.488}$$

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Capacity	Profitable.	
2	Yes.	2
2	NO	
2	NO.	
4	NO	4
4	Yes	
4	NO.	
4	NO	4.
4	Yes.	
4	Yes.	
5	Yes.	
5	Yes.	

$$\text{Gini}(\text{Capacity}(2)) = 1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2 = 0.78 \quad 0.44$$

$$\text{Gini}(\text{Capacity}(4)) = 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 = 0.5$$

$$\text{Gini}(\text{Capacity}(5)) = 1 - \left(\frac{0}{2}\right)^2 - \left(\frac{2}{2}\right)^2 = 0.$$

$$\text{Gini Indp}(\text{Capacity}) = \frac{3}{11} * 0.78 + \frac{6}{14} * 0.5$$

$$= 0.212 + 0.2727$$

$$= 0.4847 \quad 0.39$$

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2b

Record.	Age	Purchased (Yes/No)	
1	22	NO	23.5
2	25	Yes	
3	28	Yes	31.5
4	35	NO	37.5
5	40	Yes	

$$\text{Gini}(<23.5) = 1-\left(\frac{1}{1}\right)^2-\left(\frac{0}{1}\right)^2 = 0$$

I $$\text{Gini}(\geq 23.5) = 1-\left(\frac{3}{4}\right)^2-\left(\frac{1}{4}\right)^2 =$$

$$= 1-0.562-0.0625 = \underline{0.375}$$

$$\text{Gini Index (Split Pt (23.5))} = \frac{1}{5}*0+\frac{4}{5}*0.375 = 0.3$$

II $$\text{Gini}(<31.5) = 1-\left(\frac{2}{3}\right)^2-\left(\frac{1}{3}\right)^2$$

$$= 1-0.044-0.11 = \underline{0.45}$$

$$\text{Gini}(\geq 31.5) = 1-\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2 = \underline{0.5}$$

$$\text{Gini Index (Split Pt (31.5))} = \frac{3}{5}*0.45+\frac{2}{5}*0.5$$

$$= 0.27+0.2$$

$$= 0.47$$

III $$\text{Gini}(<37.5) = 1-\left(\frac{2}{4}\right)^2-\left(\frac{2}{4}\right)^2 = 0.5$$

$$\text{Gini}(\geq 37.5) = 0.$$

$$\text{Gini Index (Split Pt (37.5))} = \frac{4}{5}*0.5 = 0.47$$

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